Intention to Use eSanjeevani-OPD App: An Empirical Investigation Using Health Belief Model.

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***Abstract*— Digital health interventions have become an essential component of every public health system since the COVID-19 pandemic. "eSanjeevani OPD - Stay at Home OPD" is a telemedicine system that connects doctors and patients launched as part of the Ayushman Bharat project of the Indian government. This study analyses various factors affecting the intention to use eSanjeevani. A theoretical model integrating the health belief model and the theory of reasoned action was framed and empirically tested. Responses were collected using a survey questionnaire(n=248). A partial least square-structural equation modeling was used to analyze the linkages between the constructs. Perceived susceptibility and benefits were found to be the most contributing variables. Attitude has a significant mediating effect on the intention to use eSanjeevani.**

Keywords— eSanjeevani, health belief model, telehealth, digital health interventions

I. INTRODUCTION

COVID-19 has revolutionized healthcare with the rapid penetration and acceptability of various digital health interventions. Among these interventions, telemedicine gained popularity globally as it enabled virtual physician consultations during the pandemic. Digital health interventions were designed for the general public with various objectives, including targeted and untargeted client communications, personal health tracking, client-to-client communication, citizen-based reporting, providing information to clients on-demand, and client financial transactions[1]. The functionalities of digital health interventions for clients are presented in Figure 1. The guiding principles for implementing telemedicine strategies into the national health system include 1)patient centricity: implementing appropriate policies, regulations, and interventions to ensure quality, accessibility, and sustainability of telemedicine service; 2) Multisectoral and multi-disciplinary approach: Partnership between private and public sectors and collaboration of health and non-health sector expertise; 3) Strong digital governance: Accountability and confidentiality of data; 4)Equity and inclusivity: Widening the digital divide; 5)Usability: Co-designing with end users to adhere to best practices of usability and accessibility;6)Contextualisation and localization: Digital solution to be ready to be launched in local contexts[2].

"eSanjeevani OPD - Stay at Home OPD" is a telemedicine system that connects doctors and patients as part of the Ayushman Bharat project of the Indian government. The eSanjeevani is reportedly India's first national project in the field of digital health delivery. The program facilitated clinical consultations between a doctor at the health center and a patient at home via voice and video. Initially, patients were required to connect to the program using a web-based browser on a laptop, but after a few weeks, an android-based mobile app was created, allowing patients to join as well. Numerous difficulties raised by the COVID-19 outbreak were addressed through eSanjeevani [3]. The screenshots of the mobile App of eSanjeevani OPD are presented in figure 2.

To understand the usability and to identify the targets of the improvisation of telemedicine services, we need to understand the user perceptions of using the intervention. We identified studies that analyzed user perceptions of various digital health interventions, tabulated in Table 1. Further, regular monitoring and evaluation of digital health systems are essential. Hence, this study aims to understand user perceptions on the uptake of eSanjeevani OPD service. The well-established health belief model(HBM) and theory of reasoned action(TRA) are integrated to understand the intention to use the eSanjeevani mobile app. Thus, the objectives are

1. What are the effects of perceived barriers and benefits, external and internal cues of action, and the threat of COVID-19 on attitude towards using the eSanjeevani app?
2. What is the effect of attitude on an individual's intention to use the eSanjeevani app?

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Table 1 Selected studies on user acceptance of telemedicine programs

Reference	*Country*	*Theory*	*Constructs used*
Tsai, 2014 [4]	Taiwan	Technology acceptance model (TAM) and HBM	Social and institutional trust, perceived ease of use and usefulness, susceptibility, severity, benefits, and barriers
Rho et al., 2015 [5]	South Korea	Unified theory of acceptance and use of technology (UTUAT)	Performance and effort expectancy, facilitating conditions, Social influence
Singh & Dev, 2021 [6]	India	TAM	Perceived ease of use and usefulness, attitude,
Wu et al., 2021 [7]	China	Social cognitive theory	Motivation, Subjective norms, self-related health, self-efficacy, Institution size, Trust
Ramírez-Rivas et al., 2021 [8]	Brazil	Theory of reasoned action	Plasticity, Subjective norm, Attitude, Perceived behavior control

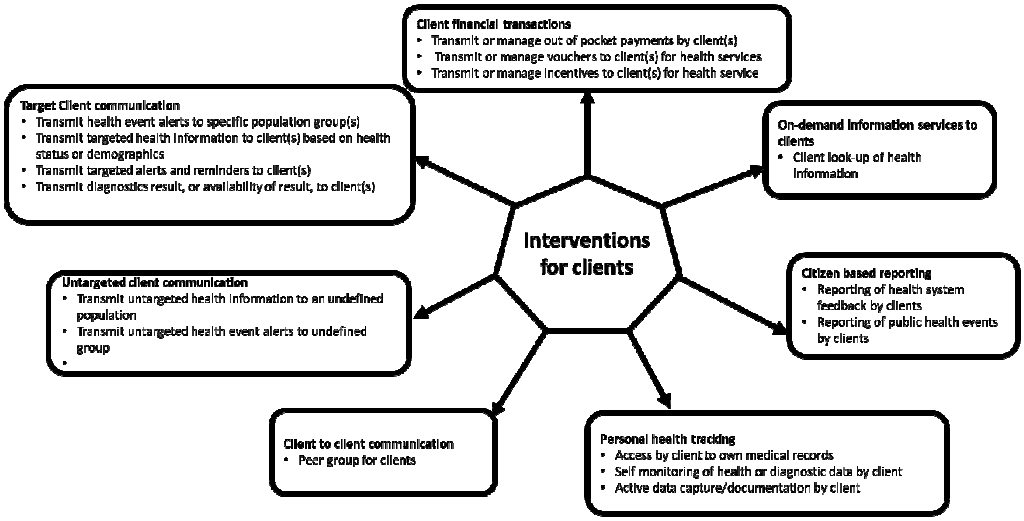


Fig. 1 Functionalities of digital health interventions for clients

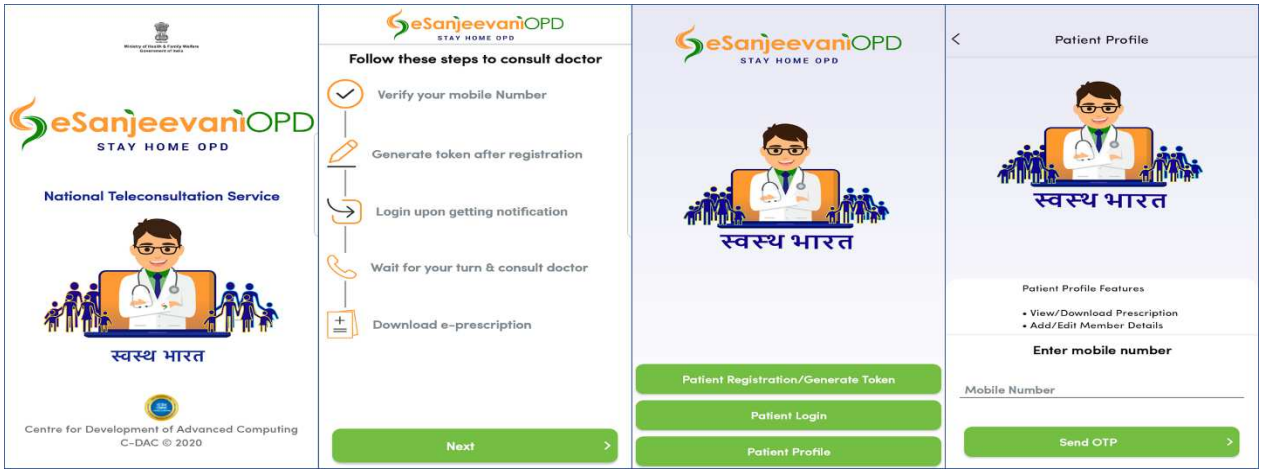


Fig. 2 Screenshots of the eSanjeevani mobile app

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II. THEORETICAL UNDERPINNINGS AND HYPOTHESIS DEVELOPMENT

Two theories were considered in this study: The health belief model(HBM) and theory of reasoned action(TRA).

HBM aims to identify why people fail to adopt disease prevention strategies or early detection screening tests. According to HBM, the two components of health-related behavior are avoidance of illness and belief that a specific health action will cure or prevent illness. The perceptions concerning the benefits and barriers were added when the theory evolved. The constructs used in HBM are perceived susceptibility, severity, barriers and benefits, and cues to action[9]. In this study, PS is associated with people's perceptions concerning the threat of COVID-19. We presume that if individuals feel that they are more worried or afraid concerning the probability of getting infected with the virus, more he will have a favorable attitude and choose virtual consulting applications like eSanjeevani. We also presume that when people are aware that using eSanjeevani will have the benefits of improved access to doctors and can reduce the chances of getting contacted with a COVID-infected person, they are more likely to develop a favorable attitude toward using the App. The platform's accessibility, user-friendly design, ease of use, and internet connectivity can make people use it. But if there are barriers like poor internet connectivity, or if the users perceive that it is difficult to use the application, they will develop an unfavorable attitude towards the App. When people receive internal cues like fever, cough, etc., they are more likely to seek healthcare consultation and thus will have a positive attitude towards using the eSanjeevani app. Further, social research has proved that social influences strongly affect attitude. So, we presume that people are more likely to have a positive attitude when their friends or family use or suggest using the eSanjeevani app

The relationship between a person's attitude and intention is explained in TRA. TRA is conceptualized considering that the action is under volitional control. Analyzing individuals' preexisting attitudes and intentions predict their behavior. Individuals engage in certain behaviors based on the outcomes they expect to experience from the behavior. Behavior intention is considered as a function of attitude. The strength of an intention is indicated by the subjective probability that a person will perform a behavior[10]. In this study, we presume that once an individual develops a positive attitude about the eSanjeevani app, the stronger will be his/her intention to use the App (IU).

III. METHODOLOGY

A. The conceptual model

Figure 3 indicates all the hypotheses considered in the study. PS, PBen, Pbar, ICA, and ECA are considered as antecedents of ATT, and ATT has a direct linkage to IU

B. Measurement Model

The indicators to measure constructs were finalized after a thorough literature review and expert consultations. The operational definitions of constructs were formulated based on contextual factors. The items were phrased into statements to use for data collection. It was content validated with experts, pilot tested, and alterations were made to eliminate all the ambiguities related to response biases. The operational definitions and the items used for the study are presented in table 1.

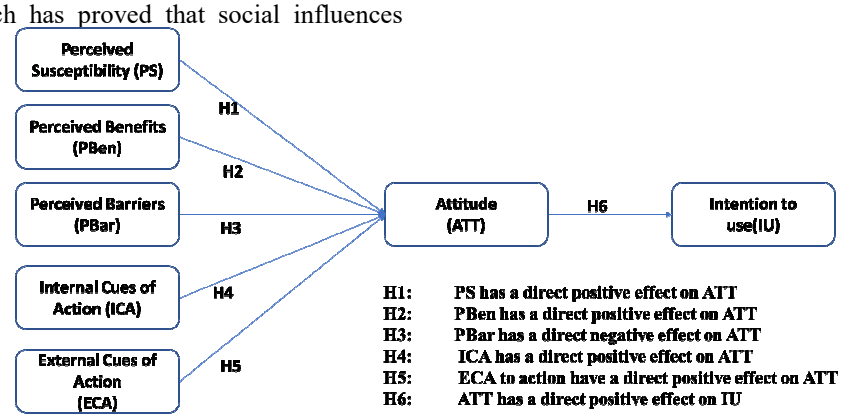


Fig. 2. The conceptual model

C. Data collection and analysis

Responses were sought using a structured questionnaire prepared in English, and it constituted two parts: Part A constituted questions on demographic characteristics, and Part B consisted of closed-ended questions related to the constructs for measurement on a five-point Likert scale which measured the level of agreements (Strongly disagree- Strongly agree) An online survey method was used. The link containing the online survey form was forwarded to the potential respondents through email and Whatsapp. Besides, we posted the link on the walls of social media pages to receive responses. All participants were informed of the goal of this study endeavor at the beginning of the questionnaire. The data collection phase of the study began in May 2021 and concluded in July 2021.248 responses were received. The sample size was adequate according to the apriori sample

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calculator for structural equation modeling (SEM). Data quality was ensured by checking missing values, outliers, normality, randomness, data independence etc. Descriptive analysis of the data was performed and quality of measurement was checked. A pooled exploratory factor analysis indicated that the items are loaded in the presumed manner.The partial least square (PLS-SEM), a variance-based WarpPLS 8.0 software, was used to analyze the data because of its ability to test linear and non-linear relationships. Subsequent estimation confirmed the requirement of all primary quality criteria and the model's fitness for hypothesis testing[11], which are presented in fig 4. The estimated model is shown in the figure 4.

TABLE 2. Scale items used for the study

Construct	*Items*	*Reference*
PS	- greater likelihood of getting COVID-19 - chance of exposure to COVID-19 - More likely to catch COVID-19 in a lifetime - Strong possibility of an attack of COVID 19 because of comorbidities - Family members have more likelihood of getting COVID 19	[12]
PBen	- Improving access to healthcare service - comfortable using the app - less time-consuming than a hospital visit - to be less concerned about the health - active involvement in health	[13]
PBar	- Technical issues while using the app - No internet connectivity - unsatisfactory voice and video quality - less ease of use - Privacy concern	[14]
ICA	- Symptoms of COVID - Worsening of chronic illness - Feeling fatigued	[15]
ECA	- Friends and family use the app - Friends and family encourage me to use - Friends and family will accept if I use eSanjeevani	[15]
ATT	- Like the App - App is convenient - Feeling positive	[16]
IU	- Will use the app - Will recommend it to others - Will educate others to use it	[17]

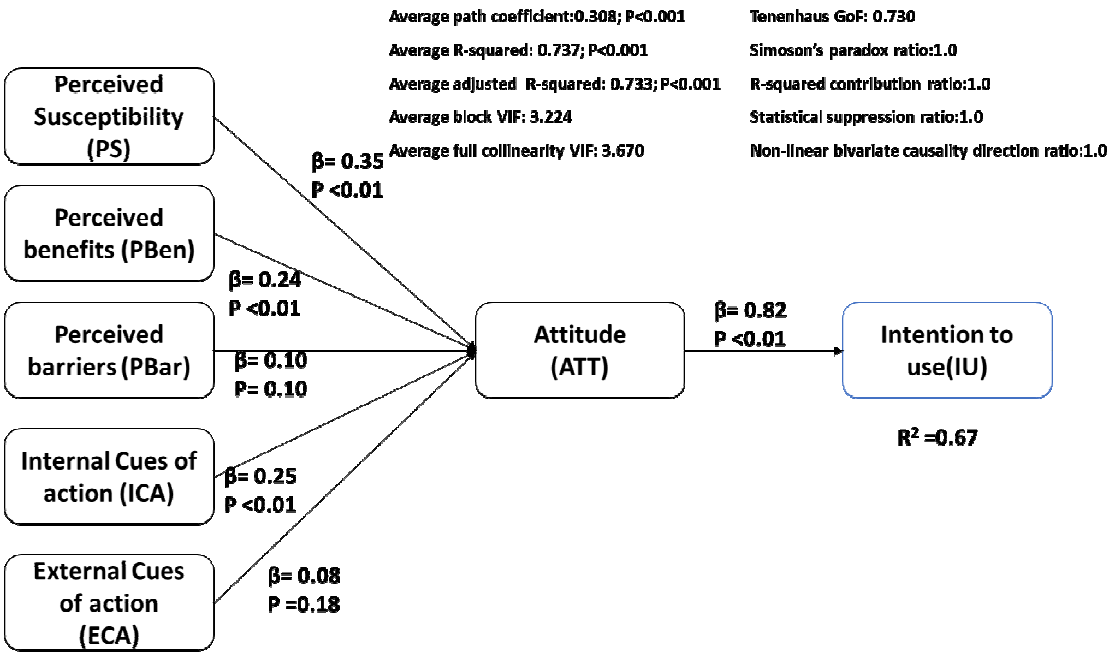


Fig.4. The estimated Model

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IV. RESULTS AND DISCUSSION

In this section, we present the results of the hypothesis as direct and indirect effects. The pathways from the predictor variables to the outcome variables are direct effects and the pathways through mediators are indirect paths. An effect on another variable is determined by a path coefficient, which represents a cause-and-effect relationship. It suggests that a relationship exists between the two variables[18]. The results are presented in table 3. First, in the case of direct effects on ATT, it is observed that all the antecedents except PBar and ECA have a significant positive effect on ATT. Among them, PS had the highest path coefficient. This indicates that when people feel afraid or concerned about the threat of COVID-19, they are more likely to have a positive attitude towards eSanjeevani. A similar effect is observed for PBen and ICA. Thus it can be concluded that once people are aware of the benefits of using the app, they are more likely to do a teleconsultation. Second, a similar trend of direct effects was observed for the indirect effects. PS, PBen, and ICA significantly predicted IU with ATT as a mediator. Additionally, ECU was significantly predicting IU in the presence of ATT. This indicates that a positive or favorable attitude is generated by social influence. People are more likely to use eSanjeevani if their friends and family have a positive opinion about the use.

From the above results, we propose the below recommendations to encourage the use of telehealth applications. The implications are not confined to managing health services during the COVID-19 pandemic, as it can be made practical during any similar public health emergencies.

a. User experience: Creating a user-friendly interface, reducing the waiting time for consultation, availability of doctors, etc., can increase the likelihood that people will use telehealth applications.

b. Awareness: During any health emergency, it is essential to create awareness concerning the severity and high probability of getting affected by the disease. The campaigns must also target to create a notion among the public about the benefits of using telehealth platforms.

c. Nudging: Frequent reminders, nudging messages through SMS and social media, and flyers in the health centers will remind the people about the virtual consulting options available for them, which can motivate them to use telehealth platforms when in need.

TABLE 3. RESULTS OF DIRECT AND INDIRECT EFFECTS EFFECTS

Direct Effects				Indirect effects			
Hypothesis	***'β' value***	***'p-value***	***Significance***	***Hypothesis***	***'β' value***	***p-value***	***Significance***
H1: PS→ ATT	0.35	<0.01	Yes	PS→IU	0.28	<0.05	Yes
H2: Pben→ATT	0.24	<0.01	Yes	PBen→IU	0.19	<0.05	Yes
H3: PBar→ATT	0.10	0.10	No	PBar→IU	0.08	0.13	No
H4: ICA→ATT	0.25	<0.01	Yes	ICA→IU	0.20	<0.01	Yes
H5: ECA→ATT	0.08	0.18	No	ECA→IU	0.06	<0.05	Yes
H6: ATT →IU	0.82	<0.01	Yes				

V. LIMITATIONS, FUTURE DIRECTIONS, AND CONCLUSION

The study empirically examined various factors influencing the intention to use the eSanjeevani app in the backdrop of the health belief model. From the analysis, it is concluded that the perceived susceptibility to the existing situation (health emergency) can create a positive attitude about eSanjeevani, and the stronger the perceived susceptibility stronger the intention to use. We hope that the implication of our study can help the actors of the health system to design and implement strategies encouraging the use of telehealth platforms. Despite the contribution, our study has limitations. India is a country with diverse cultures with different beliefs about health and medicine. We doubt whether our sample have represented all the cultural subgroups. Further, it is recommended to direct the research to understand various other factors like user experience, financial coverage of telemedicine, technology acceptance, digital divide, etc, in the context of various digital health interventions.

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